

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 14: Equation of a Line

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3, 3)$, $B(0, -2)$, $C(-4, -2)$ and $D(-1, 3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC .

ICSE 2025, 5 marks

2. Triangle ABC has vertices $A(1, -1)$, $B(0, 4)$ and $C(-6, 4)$, and D is the mid-point of BC . Find (a) the coordinates of D (b) the equation of the median AD . ICSE 2023, 3 marks

3. $A(2, 5)$, $B(-1, 2)$ and $C(5, 8)$ are the vertices of a triangle. M is the point on AB with $AM : MB = 1 : 2$. Find M , and then the equation of the line through C and M . ICSE 2018, 4 marks

2 Lines parallel or perpendicular to a given line

4. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ ICSE 2025, 1 mark
5. The line PQ meets the positive x -axis at P , 5 units from the origin, and the negative y -axis at Q , 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ . Find (a) the coordinates of P and Q (b) the equation of RS . ICSE 2025, 3 marks
6. (a) The lines $kx - y + 4 = 0$ and $2y = 6x + 7$ are perpendicular. Find k . (b) Find the equation of the line parallel to $2y = 6x + 7$ that passes through $(-1, 1)$. ICSE 2024, 3 marks
7. The lines $5x - 3y + 2 = 0$ and $6x - py + 7 = 0$ are perpendicular. Find p . Then find the equation of the line through $(-2, -1)$ parallel to $6x - py + 7 = 0$. ICSE 2020, 3 marks
8. The vertices of $\triangle ABC$ are $A(3, 8)$, $B(-1, 2)$ and $C(6, -6)$. Find (a) the slope of BC (b) the equation of the line through A perpendicular to BC . ICSE 2019, 3 marks
9. The lines $3x - 5y = 7$ and $4x + ay + 9 = 0$ are perpendicular. Find a . ICSE 2018, 3 marks
10. $A(-1, 3)$, $B(4, 2)$ and $C(3, -2)$ are the vertices of a triangle. Find (a) the centroid G (b) the equation of the line through G parallel to AC . ICSE 2017, 3 marks

3 Inclination and slope

11. The points $A(x, y)$, $B(3, -2)$ and $C(4, -5)$ are collinear. Write y in terms of x . (a) $3x - 11$ (b) $11 - 3x$ (c) $3x - 7$ (d) $7 - 3x$ ICSE 2024, 1 mark

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